

Link-Level Assessment of NOMA Aided Multi-hop DECT-2020 New Radio for mMTC Applications

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Abstract—Digital enhanced cordless telecommunication-2020 new radio (DECT-2020 NR) is an essential part of the fifth generation (5G) standards proposed by the international telecommunication union (ITU). With an achievable density of approximately one million connections per square km, it is a potential candidate which promises to fill in the actual gap in the wireless internet of things (IoT) market for the massive machine-type communications (mMTC). In this paper, we explore a non-orthogonal multiple access (NOMA) assisted cooperative multi-hop (MH) relaying system for DECT-2020 NR system for mMTC applications. The use of cooperative multi-hop (MH) relaying system allows a reliable long-haul communication system and further the use of NOMA scheme helps in achieving a better spectral efficiency and massive connectivity. The link level evaluation of the proposed system is carried out for the proposed NOMA aided MH DECT-2020 NR system and further error rate analysis is presented in terms of outage probability and the bit error rate (BER). Monte-Carlo simulations further corroborate the correctness of the analytical findings.

Index Terms—Decode-and-forward, digital enhanced cordless telecommunication-2020 new radio, massive machine-type communication, multi-hop, non-orthogonal multiple access.

I. INTRODUCTION

Fifth-generation (5G) communication systems aim to achieve high data rates, reliability, and energy efficiency with massive connectivity. One of the salient features of 5G communication, i.e., massive machine-type communication (mMTC), is an area where failure is not an option and could put at risk the complete automation processes, critical infrastructure, livelihoods, if not lives themselves. Digital enhanced cordless telecommunication-2020 new radio (DECT-2020 NR) as a part of 5G standards promises to fulfil all the gaps in the wireless internet of things (IoT) industry and supports one million connections per square km to support mMTC [1]–[3].

The mMTC scenario encompasses ad-hoc wireless networks, device-to-device connectivity, wireless sensor net-

works, and machine-to-machine applications, which require a sufficiently large connection density. Multi-hop (MH) communication can improve upon the reliability of the communication links to support such massive connectivity compared to the point-to-point communication links [4]. It enhances the system's performance by decreasing the transmit power at the transmitter and extending the network coverage area [5]. For such a system, the long transmission link segregates it into multiple short connections, which are further connected via the intermediate relay nodes [6], [7]. Based on the decoding procedure used at the relay node, some of the cooperative relaying schemes are decode-and-forward (DF), selective DF, and amplify-and-forward (AF) [8].

A. Related works

The performance analysis of cooperative relaying scheme aided NOMA and DECT-2020 NR specifications has been studied in [9]–[19]. The authors in [9] compare the efficacy of DECT-2020 NR specifications over the mMTC with the existing solutions regarding the supported density of nodes. The authors in [10] perform a thorough simulation to examine the DECT-2020 NR's performance for the mMTC and ultra-reliable low-latency communication (uRLLC) scenarios. The simulated results are beneficial for calibrating the physical layer performance for future system-level evaluation. Further, the authors in [11] evaluate DECT-2020 NR for compatibility with the uRLLC usage case simultaneously at the link and system levels. The performance of MH DF relaying is demonstrated in [12] for the mMTC scenario for the corresponding specifications. In [13], different approaches for relay selection, allocation of channel/frequency, and device association with relays are proposed. The angle-based process is the most optimal method for the scenario mentioned above.

The need for higher spectral efficiency for the next-generation communication systems is inevitable to cater for massive connectivity and long-haul scenarios. One of the most promising technology to enable multiple access to support the

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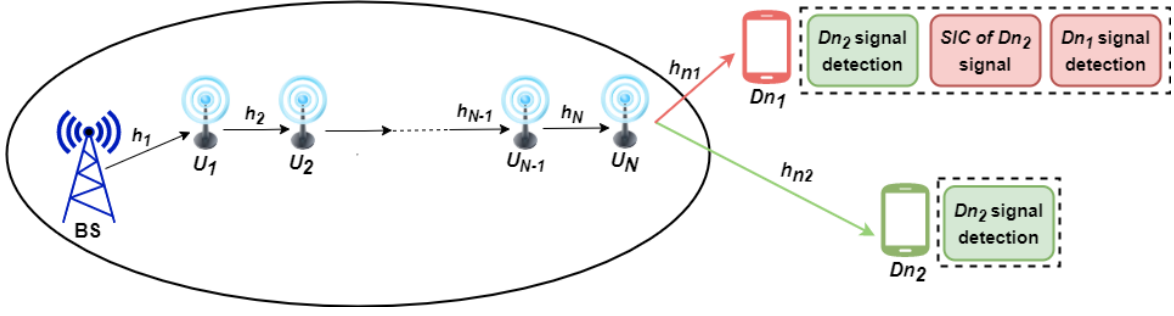


Fig. 1: A system model of considered DF MH downlink NOMA scenario

needs of the 5G networks is non-orthogonal multiple access (NOMA) [14]. The downlink NOMA uses a superimposed coding (SC) signal at the transmitter and successive interference cancellation (SIC) technique at the receiver end to detect the intended user's information. NOMA provides a higher spectrum efficiency than the conventional orthogonal multiple access (OMA) scheme at the cost of the increased error rate [15], [16]. Integrating NOMA with other physical layer techniques, including cooperative communication, MIMO system, cognitive radio networks, and visible light communication (VLC), further enhances the system's performance [17] and provides an efficient multiple access solution. Recently, significant work has been done based on the cooperative NOMA system over the point-to-point and dual-hop communication system to exploit the benefits like improved spectral efficiency and network coverage [18], [19].

B. Motivation and Contribution

DECT-2020 NR can achieve a density of approximately one million connections per square km. Hence it is a potential candidate that promises to fill the void of IoT demand for mMTC. Moreover, a cooperative MH relaying system allows a reliable long-haul communication system. And the use of the NOMA scheme helps achieve better spectral efficiency and massive connectivity. Hence, owing to its inherent merits, the MH DF NOMA-aided relaying requires more research employing DECT-2020 NR.

It inspired us to propose this system and explore its outage and bit error rate (BER) performance over static and dynamic channel environments. Furthermore, we illustrate the link-level evaluation of the system using the DECT-2020 NR specifications over various parameters. In this regard, the main contributions of the work are summarised as follows.

- The outage analysis is presented for the proposed MH DF NOMA system over the multi-path fading scenario considering Rayleigh fading channels.
- The performance of the system is presented using a varying number of hops, the distance between the nodes and the number of FFT points.
- Monte-Carlo simulations are provided to justify the correctness of the derived analysis.
- The link-level assessment of the system is also presented in terms of BER over the DECT-2020 NR specifications.

- Additionally, the performance is explored for coded and uncoded systems using such specifications.

The remaining paper is organized as follows: Section II presents the DECT 2020 NR frame structure specifications. Section III provides the considered system model of the MH downlink NOMA system. Performance analysis of such a system is presented in Section IV. Section V demonstrates the numerical results and their description. Finally, the conclusion is provided in Section VI.

II. DECT-2020 NR FRAME STRUCTURE

DECT-2020 NR is an emerging standard to meet the criteria of IMT-2020. The transmission range supported by DECT-2020 NR is 1880–1900 MHz, comprising ten varying frequency channels ranging from 1881.792 MHz to 1897.344 MHz with a bandgap of 1728 kHz [2]. Similar to NR, it also specifies a subcarrier scaling factor (μ) $\{1, 2, 4, 8\}$, resulting in a subcarrier spacing (Δf) in kHz of $\{27, 54, 108, 216\}$, respectively. Moreover, it presents a network with mesh topology having multiple radio devices connectivity via a fixed terminating (FT) point [3].

The detailed study provides the overview, transmission requirements, and physical layer specification of DECT-2020 NR specifications in the drafts released by the European telecommunications standards institute (ETSI) [1]–[3], respectively. The maximum bandwidth is 10 MHz with a symbol length of 10 *ms*. Similar to NR specifications [20], it also provides a numerology factor (μ) $= \{1, 2, 4, 8\}$. The subcarrier spacing (in kHz) and the number of orthogonal frequency-division multiplexing (OFDM) symbols per slot are provided as 10μ and 27μ , respectively. Further, the number of FFT points, a cyclic prefix (L_{CP}) and useful subcarriers are given as $\{64, 128, 256, 1024\}$, $\{8, 16, 32, 128\}$, and $\{56, 112, 224, 896\}$, respectively.

III. SYSTEM MODEL

A MH DF downlink NOMA system having one source base station (BS), two NOMA users and N intermediate users/relays is considered, as shown in Fig. 1. Based on the location, the relay U_N is assumed to be nearer to NOMA users. Since there is no direct link between the BS and NOMA users in long-haul communications, we assume that both the near user (Dn_1) and the far user (Dn_2) are beyond

the BS's communication range. The channel between BS to k^{th} intermediate user is assumed to be independent and identically distributed (i.i.d.) Rayleigh faded and distributed as $\sim \mathcal{CN}(0, 1) \forall k \in \{1, 2, \dots, N\}$.

With the relay's proximity, the channel condition between the U_N and the Dn_1 is assumed to be better as compared to the channel condition of the U_N - Dn_2 link, i.e., $|h_{n_1}|^2 > |h_{n_2}|^2$. At the BS, the SC symbol (x_{sc}) is generated by adding the appropriately scaled modulated symbols s_1 and s_2 . Further, the BS transmits the SC symbols to the users and is given as

$$x_{sc} = \sqrt{\alpha P_s} s_1 + \sqrt{(1 - \alpha) P_s} s_2, \quad (1)$$

where P_s denotes the power of the transmitted symbol, s_1 and s_2 are modulated symbols corresponding to the Dn_1 and Dn_2 user, respectively.

For simplification, we consider the system model in two parts, i.e., the first consisting of BS to N intermediate users and the second consisting of N^{th} intermediate user to NOMA users. For the first part of the considered system, the BS transmits the SC symbol to the 1^{st} intermediate relay. Further, as per the DF protocol, it decodes and re-transmits the data to the next intermediate user. For k^{th} intermediate user, the received data is decoded version of data from the previous $(k - 1)^{\text{th}}$ intermediate user and further, it re-transmits to the next $(k + 1)^{\text{th}}$ intermediate user. The received symbol at k^{th} intermediate user is given as

$$y_k = h_k \hat{x}_{U_{k-1}} + z_k, \quad (2)$$

where $\hat{x}_{U_{k-1}}$ is the received and decoded SC symbol at the $(k - 1)^{\text{th}}$ intermediate user with $\hat{x}_{U_0} = x_{sc}$, h_k is the Rayleigh distributed channel coefficient and z_k is the AWGN for k^{th} intermediate user $\sim \mathcal{CN}(0, N_0)$.

Further, detecting the signal at N^{th} intermediate user using (2), the received SC symbol at the intended NOMA user is provided as follows

$$y_{Dn_i} = h_{n_i} \hat{x}_{U_N} + z_{n_i}, \quad i = 1, 2, \quad (3)$$

where $\hat{x}_{U_N}$ is the decoded symbol at the N^{th} intermediate user, h_{n_i} is the i.i.d. Rayleigh faded channel coefficient of the i^{th} user and z_{n_i} is the AWGN of i^{th} NOMA user.

IV. OUTAGE ANALYSIS

The probability of an outage for the considered MH DF downlink NOMA system is examined in this Section. The MH relaying system experiences an outage event when at least SNR γ_k of any hop falls below a predetermined SNR (γ_{th}). Thus, the outage probability (P) of the N hops consisting N intermediate users is given as

$$P = 1 - \prod_{k=1}^N P_r(\gamma_k > \gamma_{\text{th}}), \quad (4)$$

where $\gamma_{\text{th}} = 2^{NR} - 1$, R is the transmission rate, and $P_r(\cdot)$ indicates the probability operator.

Further, (4) is simplified using the cumulative distribution function (CDF) of the random variable (RV) γ_k in [7, Eq. (4)] is given as

$$P = 1 - \prod_{k=1}^N (1 - F_{\gamma_k}(\gamma_{\text{th}})). \quad (5)$$

For multipath fading scenario between the various hops, the expression (5) is expressed as

$$P = 1 - \prod_{k=1}^N \left(1 - \exp\left(-\frac{\gamma_{\text{th}} d_k}{\bar{\Gamma}_k}\right) \right), \quad (6)$$

where d_k is the distance between the k^{th} hop and it is assumed to be unity, i.e., $d_k = d = 1$. $\bar{\Gamma}_k = \gamma_k \left(\frac{L_{ch}}{N_{\text{fft}}}\right)$ is the factor, being used for converting multipath fading into the flat fading with L_{ch} number of channel taps and N_{fft} FFT points.

The CDF of RV $\bar{\Gamma}_{n_1}$ corresponding to the link between the end intermediate user to the link NOMA user Dn_1 over the multipath fading scenario is given as

$$P^{Dn_1} = 1 - \exp\left(-\frac{\gamma_{\text{th}} dn_1}{2\bar{\Gamma}_{n_1}}\right). \quad (7)$$

The CDF of the RV $\bar{\Gamma}_{n_2}$ corresponding to the link between the end intermediate user to NOMA user Dn_2 over the multipath fading scenario is given by $\left(P_r\left(\frac{|h_{n_2}|^2 \varphi_2 \bar{\rho}_s}{|h_{n_2}|^2 \varphi_1 \bar{\rho}_s + 1} < \gamma_0\right)\right)$ in [21, Eq. (7)]. Thus, utilizing rigorous mathematics, the simplified equation is given as follows.

$$P^{Dn_2} = \begin{cases} 1 - \exp\left(-\frac{\gamma_{\text{th}} dn_2}{2\bar{\Gamma}_{n_2}}\right), & \gamma_{\text{th}} < \frac{1-\alpha}{\alpha}, \\ 1, & \text{otherwise,} \end{cases} \quad (8)$$

where $\bar{\Gamma}_{n_1} = \alpha \bar{\gamma}_n \left(\frac{L_{ch}}{N_{\text{fft}}}\right)$ and $\bar{\Gamma}_{n_2} = K \bar{\gamma}_n \left(\frac{L_{ch}}{N_{\text{fft}}}\right)$, $K = (1 - \alpha) - \alpha \gamma_{\text{th}}$. $\bar{\gamma}_n$ is the output SNR at the NOMA users. dn_1 and dn_2 are the distances between the end intermediate user to NOMA users Dn_1 and Dn_2 , respectively, which is given as $dn_1 = d$ and $dn_2 = \{1.5d, 3d\}$.

Therefore, the end-to-end outage probability at Dn_1 and Dn_2 of a MH NOMA system over the multipath fading scenario is obtained as

$$P^{BS-Dn_1} = 1 - \left\{ \prod_{k=1}^N \left(1 - \exp\left(-\frac{\gamma_{\text{th}} d_k}{\bar{\Gamma}_k}\right) \right) \right\} \times \left(1 - \exp\left(-\frac{\gamma_{\text{th}} dn_1}{2\bar{\Gamma}_{n_1}}\right) \right). \quad (9)$$

$$P^{BS-Dn_2} = 1 - \left\{ \prod_{k=1}^N \left(1 - \exp\left(-\frac{\gamma_{\text{th}} d_k}{\bar{\Gamma}_k}\right) \right) \right\} \times \left(1 - \exp\left(-\frac{\gamma_{\text{th}} dn_2}{2\bar{\Gamma}_{n_2}}\right) \right). \quad (10)$$

V. RESULTS AND DISCUSSION

The simulation results for outage probability and BER for the considered MH DF NOMA system are presented in this section. The specifications of DECT-2020 NR mentioned in Section II are utilized for the simulations. The results are also shown for different transmission rate values (R), the distance

between NOMA users and the source, and various FFT points. Additionally, employing DECT-2020 NR specifications, turbo coding is performed to show the benefits of coding gain over the uncoded system. For the simulation, α is chosen to be 0.4, and the channel variances for the near user (NU) and far user (FU) are considered 0 and -3 dB, respectively. For a multipath fading scenario, we have chosen $L_{ch} = 5$.

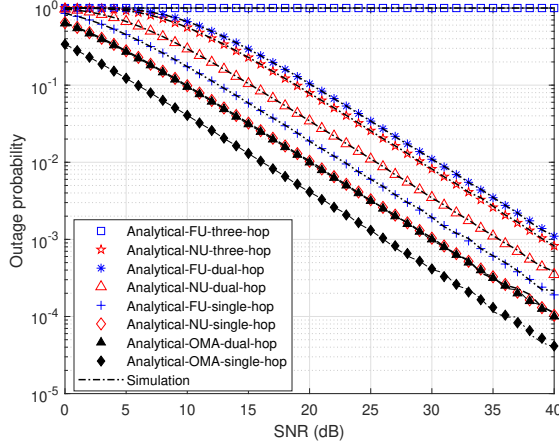


Fig. 2: Outage performance comparison of MH NOMA and OMA system

Fig. 2 presents the outage probability performance of the MH DF NOMA system using (9) and (10) for users Dn_1 and Dn_2 , respectively. The results are presented for up to three-hop scenarios over the flat fading environment considering the transmission rate $R = 0.5$ bps. A comparison between MH NOMA and OMA is demonstrated. The outage performance for the single-hop NOMA system is verified from [21]. It is noticeable that the NOMA system over single-hop outperforms the dual-hop and three-hop case as increasing the number of hops decreases the received SNR at each hop due to the fading effect. The performance of Dn_1 is better than Dn_2 due to better channel conditions. It is observed that increasing the hops beyond two degrades the overall outage performance severely. The outage performance of single and dual-hop OMA systems is better than NOMA systems due to the absence of inter-user interference at OMA users. Hence, it is noted that the considered system is used for massive connections with improved spectral efficiency over the conventional OMA system despite decreasing the outage performance.

The MH NOMA system's outage performance with the distance variation between the users is shown in Fig. 3. For simulations, the distance between intermediate users and near user is $d_k = dn_1 = d = 1$ is chosen. Similarly, the distance between far and near user is $dn_2 = \{1.5d, 3d\}$ is considered. It is observed that the performance at Dn_1 user decreases on increasing the number of hops. Also, the performance at Dn_2 decreases with increasing the distance between the end intermediate node to the Dn_2 user for single and dual-hop systems. Hence, it is noted that the proposed system can be used where the distance between the nodes varies.

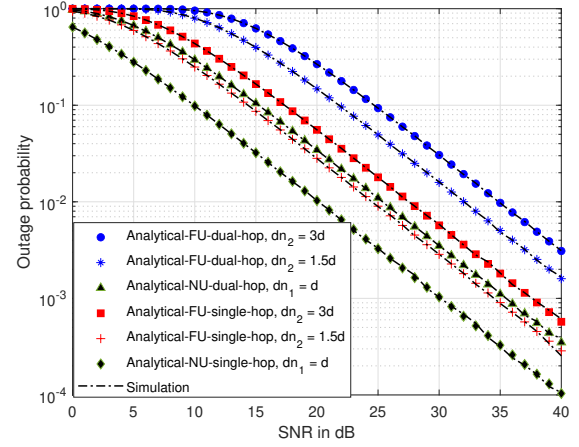


Fig. 3: Outage performance of MH NOMA system with varying distance $d_k = dn_1 = d = 1$ and $dn_2 = \{1.5d, 3d\}$

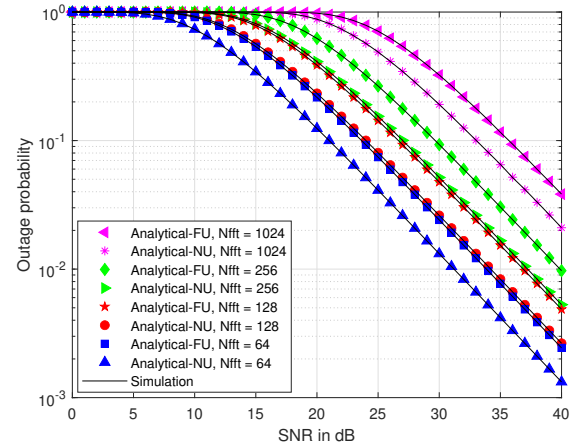


Fig. 4: Outage performance of single-hop NOMA system over multipath fading scenario

Fig. 4 represents the outage probability performance of the single-hop NOMA system over the multipath fading scenario for $\{64, 128, 256, 1024\}$ FFT points as per the DECT-2020 NR specifications [12]. It has been shown that as transmitted sub-carrier power diminishes, the chance of outages for both users increases as the FFT point increases. Hence, it is worth mentioning that the proposed system works over the multipath fading scenario for the described specifications.

Fig. 5 demonstrates the BER comparison of uncoded and coded (using convolutional turbo code (CTC)) single-hop NOMA system with code rate 1/3 as per the DECT-2020 NR specification. It is observed that the coded system enhances performance more than the uncoded system. The significant improvement of 6-7 dB can be seen at an error floor of 10^{-3} for both the NOMA users over the coded transmission.

Moreover, the BER performance comparison for the single-hop NOMA system over DECT-2020 NR specification is shown in Fig. 6 over the multipath fading scenario by

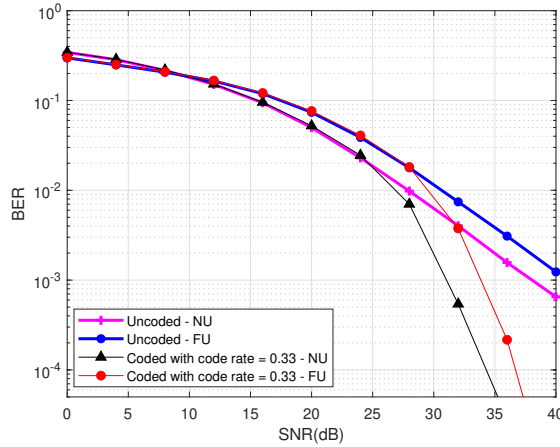


Fig. 5: Comparison of coded (CTC-1/3) and uncoded NOMA system over flat fading scenario

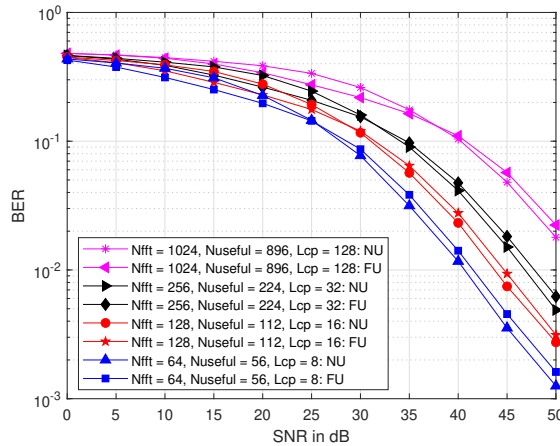


Fig. 6: Performance of NOMA system for different multipath fading scenario

varying FFT points. For simulations, the transmitted useful sub-carriers are taken $\{56, 112, 224, 896\}$ for the FFT points $\{64, 128, 256, 1024\}$, respectively. It has been shown that performance improves with fewer FFT points since adding more FFT points reduces the sub-carriers transmitted power.

VI. CONCLUSION

This paper analyses the MH relaying NOMA system over the multipath fading scenario. The system's outage performance deteriorates with increased hops, distance, and FFT points, and numerical results are corroborated with the generated simulation results. The link level performance is provided over the DECT-2020 NR specifications. The addition of 64 FFT points, 56 useful sub-carriers, and eight cyclic prefix lengths enhances the system's BER performance. Additionally, the performance of the coded system outperforms the uncoded transmission by 6-7 dB.

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